

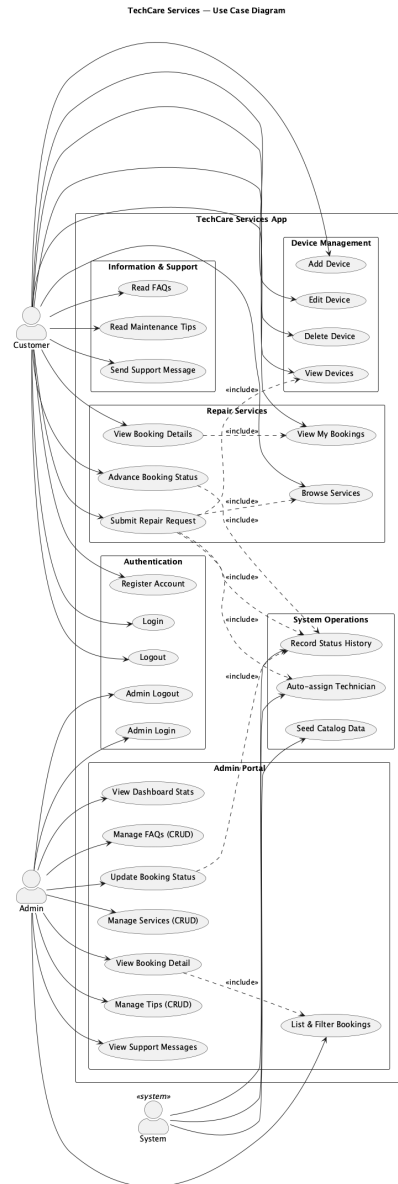
TechCare Services

System Design Documentation
Use Case · Class · Activity · ER · Relational Schema

Android repair-booking application (MVVM + Room/SQLite). Database file: **techcare.db** (version 4).

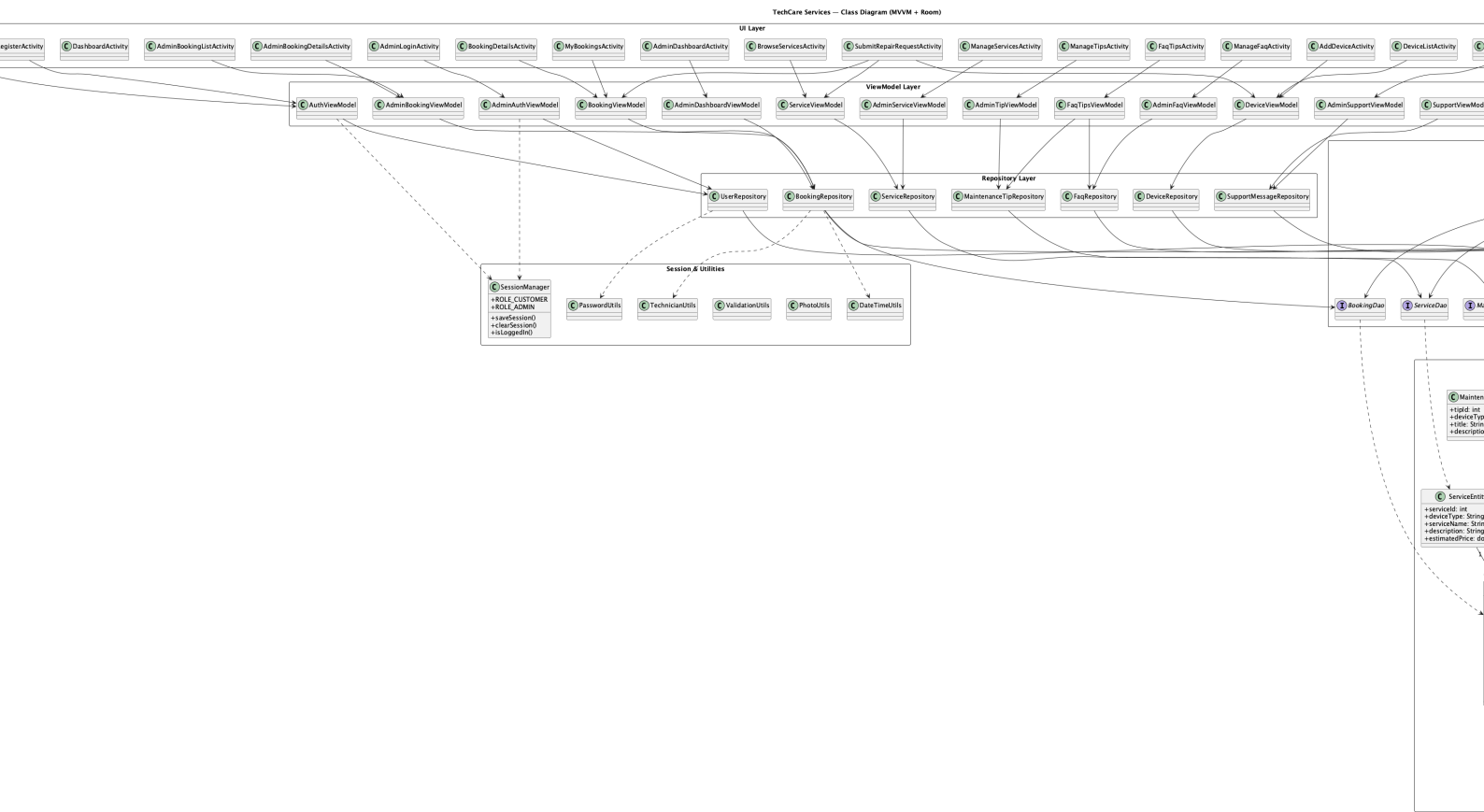
1. Use Case Diagram

Actors: **Customer**, **Admin**, and **System**. Customers register devices, browse services, submit repair requests, track bookings, and contact support. Admins manage bookings, catalog content, and support messages.



2. Class Diagram

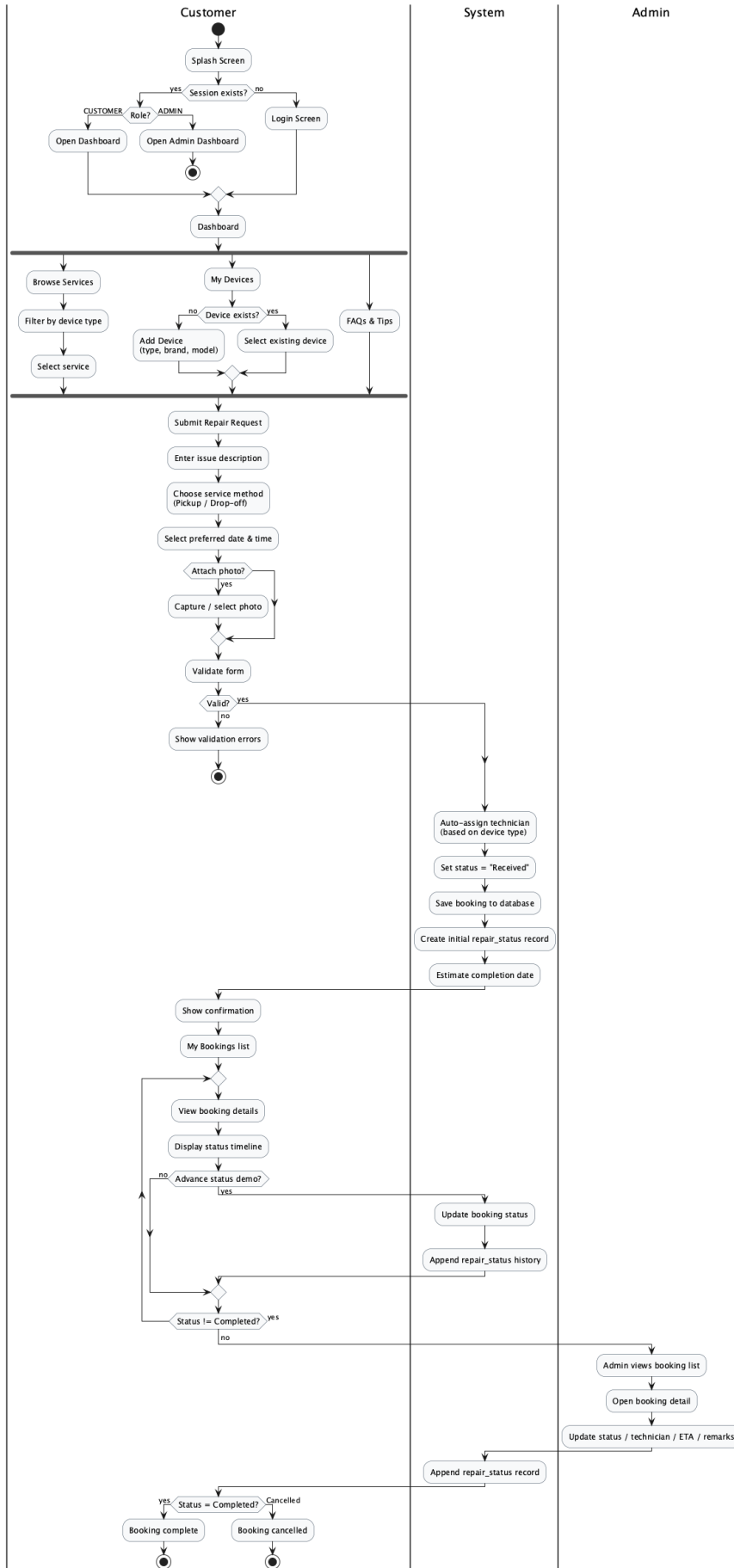
MVVM architecture: Activities observe ViewModels, which delegate to Repositories and Room DAOs. Entities map to SQLite tables in AppDatabase.



3. Activity Diagram

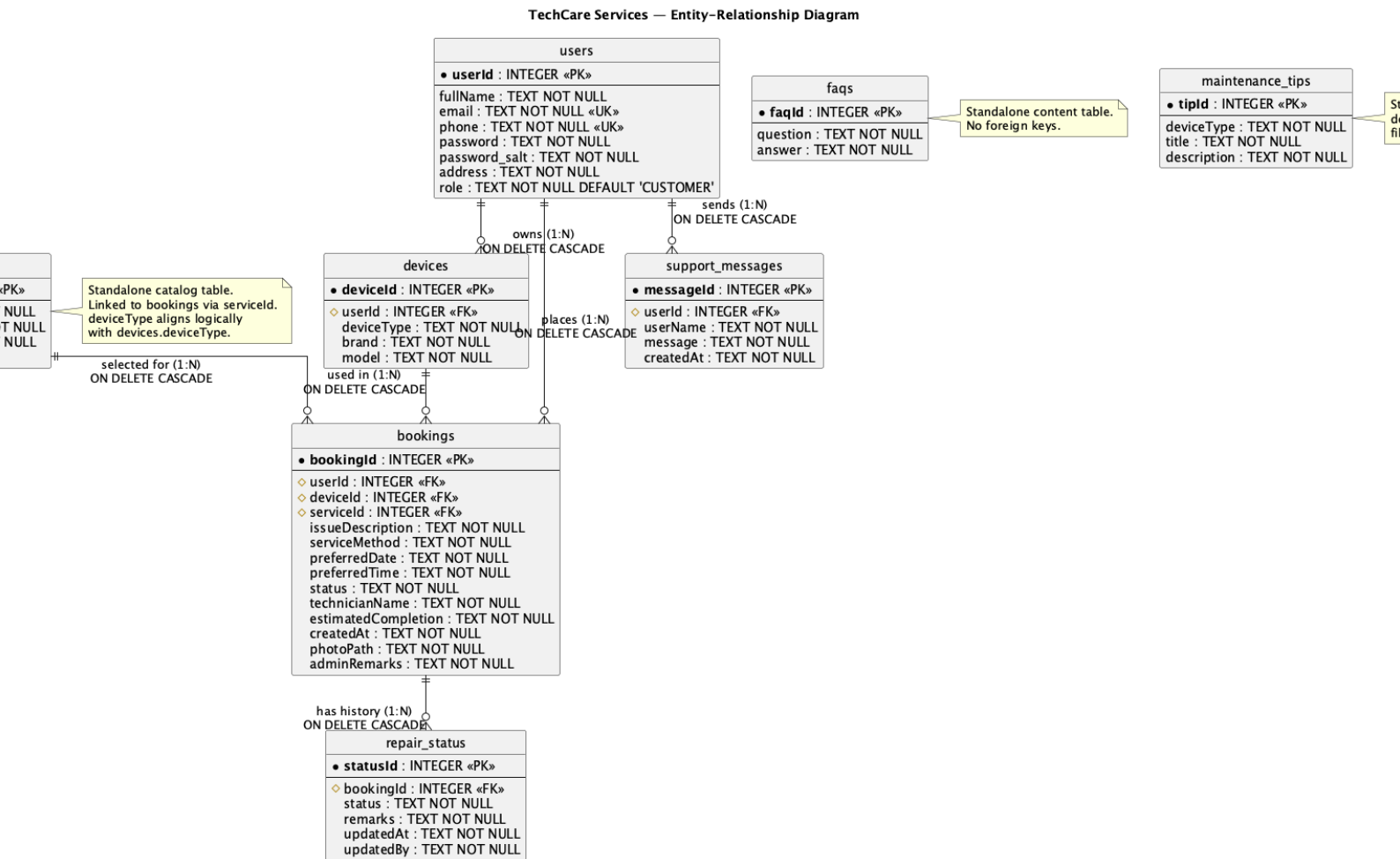
End-to-end repair booking flow from splash/login through submission, status tracking, and admin updates.

TechCare Services — Repair Booking Activity Diagram



4. Entity-Relationship Diagram

Eight tables with foreign-key relationships. bookings is the central fact table joining users, devices, and services. repair_status stores the booking status timeline.



5. Normalized Relational Schema

The schema below is in **Third Normal Form (3NF)**: each non-key attribute depends only on the primary key, and there are no transitive dependencies. Referential integrity uses ON DELETE CASCADE on all foreign keys.

Relationship Summary

Relationship	Cardinality	FK / Rule
users → devices	1 : N	devices.userId → users.userId
users → bookings	1 : N	bookings.userId → users.userId
users → support_messages	1 : N	support_messages.userId → users.userId
devices → bookings	1 : N	bookings.deviceId → devices.deviceId
services → bookings	1 : N	bookings.serviceId → services.serviceId
bookings → repair_status	1 : N	repair_status.bookingId → bookings.bookingId

Table: users

3NF — No transitive dependencies; role is functionally dependent on userId.

Column	Type	Constraints
userId	INTEGER	PRIMARY KEY, AUTOINCREMENT
fullName	TEXT	NOT NULL
email	TEXT	NOT NULL, UNIQUE
phone	TEXT	NOT NULL, UNIQUE
password	TEXT	NOT NULL (PBKDF2 hash)
password_salt	TEXT	NOT NULL
address	TEXT	NOT NULL
role	TEXT	NOT NULL, DEFAULT 'CUSTOMER'

Table: devices

3NF — deviceType/brand/model depend only on deviceId; userId is FK only.

Column	Type	Constraints
deviceId	INTEGER	PRIMARY KEY, AUTOINCREMENT
userId	INTEGER	NOT NULL, FK → users(userId) ON DELETE CASCADE
deviceType	TEXT	NOT NULL
brand	TEXT	NOT NULL
model	TEXT	NOT NULL

Table: services

3NF — Catalog entity; all attributes depend on serviceId.

Column	Type	Constraints
serviceId	INTEGER	PRIMARY KEY, AUTOINCREMENT
deviceType	TEXT	NOT NULL
serviceName	TEXT	NOT NULL
description	TEXT	NOT NULL

Column	Type	Constraints
estimatedPrice	REAL	

Table: bookings

3NF — Booking facts reference users/devices/services via FKs; no redundant user data.

Column	Type	Constraints
bookingId	INTEGER	PRIMARY KEY, AUTOINCREMENT
userId	INTEGER	NOT NULL, FK → users(userId) ON DELETE CASCADE
deviceId	INTEGER	NOT NULL, FK → devices(deviceId) ON DELETE CASCADE
serviceId	INTEGER	NOT NULL, FK → services(serviceId) ON DELETE CASCADE
issueDescription	TEXT	NOT NULL
serviceMethod	TEXT	NOT NULL
preferredDate	TEXT	NOT NULL
preferredTime	TEXT	NOT NULL
status	TEXT	NOT NULL
technicianName	TEXT	NOT NULL
estimatedCompletion	TEXT	NOT NULL
createdAt	TEXT	NOT NULL
photoPath	TEXT	NOT NULL, DEFAULT ''
adminRemarks	TEXT	NOT NULL, DEFAULT ''

Table: repair_status

3NF — Status history normalized; each event depends on statusId.

Column	Type	Constraints
statusId	INTEGER	PRIMARY KEY, AUTOINCREMENT
bookingId	INTEGER	NOT NULL, FK → bookings(bookingId) ON DELETE CASCADE
status	TEXT	NOT NULL
remarks	TEXT	NOT NULL
updatedAt	TEXT	NOT NULL
updatedBy	TEXT	NOT NULL, DEFAULT 'System'

Table: faqs

3NF — Independent FAQ content; question/answer depend on faqId.

Column	Type	Constraints
faqId	INTEGER	PRIMARY KEY, AUTOINCREMENT
question	TEXT	NOT NULL
answer	TEXT	NOT NULL

Table: maintenance_tips

3NF — Tips content normalized per tipId.

Column	Type	Constraints
tipId	INTEGER	PRIMARY KEY, AUTOINCREMENT
deviceType	TEXT	NOT NULL

Column	Type	Constraints
title	TEXT	NOT NULL
description	TEXT	NOT NULL

Table: support_messages
3NF — userName denormalized for display; message facts keyed by messageId.

Column	Type	Constraints
messageId	INTEGER	PRIMARY KEY, AUTOINCREMENT
userId	INTEGER	NOT NULL, FK → users(userId) ON DELETE CASCADE
userName	TEXT	NOT NULL
message	TEXT	NOT NULL
createdAt	TEXT	NOT NULL